CodeRanger: A Zero Knowledge Cross Device Password Management System for Android

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***Abstract*—With the increasing use of digital services, users today have to manage multiple passwords and other sensitive information. However, most existing password managers either depend on server-side trust or do not provide strong security across devices.**

We present CodeRanger, a zero-knowledge Android password manager that uses strong encryption, biometric authentication, and secure cloud synchronization. In each vault entry we encrypted using a separate key, ensuring that the data remains protected even if the cloud server is compromised. The application also includes machine learning features for password strength checking and detecting unusual activity, adding an extra layer of security.

***Index Terms*—Zero Knowledge Encryption, Android Security, AES GCM, Password Manager, Biometric Authentication, PBKDF2, HMAC, Secure Cloud Synchronisation, Machine Learning, Mobile Cryptography.**

I. INTRODUCTION

The rapid growth of digital services has significantly increased our dependence on password-based authentication. However, users often struggle to remember multiple unique passwords and therefore use insecure storage methods or reuse the same password across different platforms. Similarly, physical access codes for lockers and padlocks also create security challenges that need proper management.

CodeRanger addresses these issues through a cryptographically secure Android application developed using Java and Android Studio. The system provides secure local storage along with optional cross-device synchronisation using zero-knowledge encryption. This means the server never has access to plaintext credentials or encryption keys. As a result, even if the cloud infrastructure is compromised, the user's credentials remain protected.

II. RELATED WORK

Password managers have been extensively evaluated from perspectives of cryptographic security, usability, synchronisation models, and trust assumptions. However existing systems continue to exhibit the structural weaknesses in authenticated encryptions, key isolation, and zero knowledge guarantees.

Subev and Petrov [1] had conducted an in depth forensic and reverse engineering analysis of Android password managers, revealing that improper authenticated encryption usage, metadata manipulation vulnerabilities, and decrypted artefact leakage in local storage. Their work demonstrates that confidentiality without strict integrity protection leaves systems vulnerable to the tampering attacks. CodeRanger mitigates this limitation through per entry AES 256 GCM authenticated encryption that is combined with hierarchical key isolation, ensuring confidentiality and authenticity under chosen ciphertext attack models.

Kalra and Patel [2] implemented a web based encrypted password manager using Python and MySQL. Although encryption was incorporated the system lacked zero knowledge guarantees and relied on server side trust assumptions. Additionally, no per entry key compartmentalisation was implemented. In contrast CodeRanger derives cryptographic material exclusively client side using PBKDF2 and expands keys hierarchically:

$$K_d = \text{HMAC}(K, \text{DeviceID}), \quad K_e = \text{HMAC}(K_d, \text{EntryID})$$

thereby isolating compromise impact at device and record granularity.

Sharma and Mishra [3] performed a practical security analysis of Android password managers and identified weak random generation mechanisms and insecure metadata persistence. However their work remains evaluative rather than architectural. CodeRanger incorporates secure random generation compliant with NIST entropy guidelines, and integrates authenticated metadata binding.

Oesch and Ruoti [4] evaluated 13 popular password managers and discovered weaknesses in generation entropy and autofill security. Their findings emphasise implementation inconsistencies across platforms. CodeRanger avoids browser autofill attack vectors entirely, by confining execution within Android sandbox boundaries.

Duan et al. [5] formally analysed master password based password management protocols and demonstrated susceptibility to offline guessing under weak derivation schemes. CodeRanger mitigates such risks using high iteration PBKDF2 combined with salt based derivation, and hardware backed keystore storage.

Li and Li [6] provided a comprehensive taxonomy of password managers (local, cloud, hybrid) but did not propose cryptographic enhancements to eliminate centralised trust. CodeRanger adopts a zero knowledge synchronisation model

where encrypted payloads are transmitted without exposing key material to backend infrastructure.

Horsch et al. [7] proposed deterministic password derivation without centralised storage. While eliminating stored credentials deterministic derivation limits support for heterogeneous secret types such as encrypted credit card metadata and secure notes. CodeRanger preserves encrypted storage flexibility, while maintaining zero knowledge synchronisation.

Gupta et al. [8], Wang et al. [9], and Zimmermann et al. [10] evaluated password strength metres and demonstrated limitations of entropy only models. These works show that static estimators fail to capture behavioural anomalies. CodeRanger extends strength evaluation using entropy computation:

$$H = L \cdot \log_2(N)$$

combined with logistic regression classification, and lightweight autoencoder based anomaly detection.

Park et al. [11] studied biometric authentication in password managers and concluded that biometrics improve usability but do not inherently strengthen storage security. CodeRanger integrates biometric gating with hardware backed Android Keystore protection ensuring biometric authentication unlocks cryptographic keys rather than merely user interface access.

Johnson and Garcia [12] examined trust perceptions in password managers and identified server compromise concerns as a primary adoption barrier. CodeRanger directly addresses this through a strict zero knowledge architecture, that eliminates plaintext server exposure entirely.

Collectively, existing literature highlights recurring deficiencies in authenticated encryption enforcement, key compartmentalisation, and adaptive intelligence integration. CodeRanger advances beyond prior systems by combining hierarchical HMAC based key derivation, AES 256 GCM per entry authenticated encryption, hardware backed keystore protection, zero knowledge cross device synchronisation, and integrated ML based strength and anomaly analysis within a unified Android architecture.

III. COMPARATIVE ANALYSIS

IV. OBJECTIVES

The primary objectives of this research work are as follows:

- Design a zero knowledge encrypted credential management system
- Implement authenticated encryption which has hierarchical key derivation
- Integrate biometric and master password authentication mechanisms
- Enable encrypted cross device synchronisation so that users can use multiple android devices
- Incorporate the machine learning techniques for password strength evaluation

V. SCOPE

The system supports storage of various types of sensitive information:

- Website and application credentials
- Credit and debit card information (excluding CVV)
- Physical lock combinations and access codes
- Secure notes and WiFi credentials

CodeRanger operates offline by default with optional encrypted cloud synchronisation.

VI. CRYPTOGRAPHIC ARCHITECTURE

A. Key Derivation

Given a user master password P and randomly generated salt s, we derive the primary encryption key as the following:

$$K = \text{PBKDF2}(P, s, 100000, 256) \tag{1}$$

The iteration count of 100,000 renders resistance against brute force attacks, while remaining practical for mobile devices.

B. Hierarchical Key Expansion

To isolate credentials across devices and entries we implement hierarchical key derivation. Each device receives a unique key:

$$K_d = \text{HMAC}(K, \text{DeviceID}) \tag{2}$$

Each vault entry is encrypted with its own key:

$$K_e = \text{HMAC}(K_d, \text{EntryID}) \tag{3}$$

This architecture ensures that compromising one entry does not expose others, thereby maintaining data compartmentalisation.

C. Authenticated Encryption

Each credential entry M is encrypted using AES 256 GCM:

$$C, T = \text{AES GCM}(K_e, \text{IV}, M) \tag{4}$$

where IV is a 96 bit random nonce, C represents the ciphertext, and T is the authentication tag. The stored format is: $\{\text{IV}, C, T\}$. This construction provides both confidentiality and integrity protection against chosen ciphertext attacks.

VII. MACHINE LEARNING COMPONENTS

A. Password Strength Estimation

We estimate password entropy using the following formula:

$$H = L \cdot \log_2(N) \tag{5}$$

where L is password length and N represents character set size. A logistic regression classifier evaluates password strength based on entropy, dictionary proximity, and pattern repetition. This provides users with immediate feedback on password quality, thereby encouraging better password hygiene.

B. Anomaly Detection

A lightweight autoencoder monitors user behaviour including the following parameters:

- Login time patterns
- Access frequency distribution
- Device fingerprint changes

When reconstruction error exceeds threshold ε the system flags potentially suspicious activity.

TABLE I
COMPARISON OF EXISTING SYSTEMS WITH CODERANGER

Work	Zero Knowledge	Per Entry Key Isolation	Authenticated Encryption	ML Integration
Subev [1]	No	No	Partial	No
Kalra [2]	No	No	Basic AES	No
Sharma [3]	No	No	Limited	No
Oesch [4]	No	No	Varies	No
Duan [5]	Protocol Level	No	Formal Analysis	No
Li [6]	Survey			No
Horsch [7]	Yes	Deterministic	Yes	No
Gupta [8]	No	No		Strength Only
Park [11]	No	No	No	No
Johnson [12]	Conceptual	No	No	No
CodeRanger	Yes	Yes (HMAC based)	AES 256 GCM	Yes

VIII. SYSTEM ARCHITECTURE

The system architecture is divided into following layers:

Frontend Layer:

- Android Studio (Java)
- Material Design 3 UI components
- BiometricPrompt API integration

Storage Layer:

- Room Database for local persistence
- Encrypted blob storage

Cloud Synchronisation Layer:

- Firebase Authentication
- Firestore for encrypted payload storage

The zero knowledge architecture ensures that cloud servers never receive plaintext data or encryption keys whatsoever.

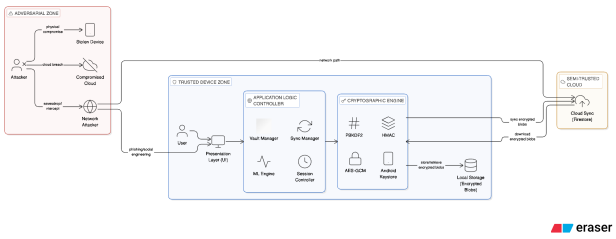

Fig. 1. Zero-Knowledge Encryption and Synchronization Architecture

IX. HARDWARE REQUIREMENTS

Development Environment:

- Intel i5 or Apple M1 processor (or equivalent)
- Minimum 8GB RAM
- Android Studio Arctic Fox or later

Target Device:

- Android 8.0 or higher
- Biometric authentication hardware
- Hardware backed keystore support

X. KEY FEATURES

The system provides following key features:

- Master password authentication with PBKDF2 key derivation
- Biometric unlock using device hardware
- Automatic locking after configurable inactivity period
- Cross device encrypted synchronisation
- Automatic clipboard clearing after timeout
- Category based vault organisation
- Dark mode interface

XI. SECURITY MODEL

The threat model considers following attack vectors:

- Physical device theft or loss
- Cloud database compromise
- Brute force password attacks
- Replay and tampering attacks

Our security guarantees include:

- Strong key derivation with 100,000 iterations
- Per entry encryption keys preventing cross contamination
- Authenticated encryption preventing tampering
- Zero plaintext storage on device or cloud

XII. CONCLUSION

CodeRanger presents a cryptographically rigorous password management system for Android, that balances security with usability. The zero knowledge architecture ensures that even complete server compromise cannot expose user credentials. Notably hierarchical key derivation provides isolation between entries while authenticated encryption prevents tampering. Biometric authentication improves usability without weakening security and machine learning components help users maintain strong passwords. The system provides a pragmatic solution for secure credential management in adversarial environments, making it suitable for real world deployment.

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